Introduction to Databases

TU257

Lesson Outline

- What is a database
- Why use databases (Databases vs. Spreasheets)
- Types of Databases
- DBMS
- Database Design and models
- Software

What is a database?

- A **database** is a structured collection of records.
  - ***Database:***
    - Collection of related data
      - Represents some aspect of the real world.
      - Logically coherent collection.
      - Built with a specific purpose in mind.
      - Could be anywhere - notebook, spreadsheet, Access, Oracle, … could be manual/could be computerised.

What is a database?

- A **database** is a structured collection of records.
  - ***Data:***
    - Raw facts (building blocks of information).
    - Unprocessed information.
  - ***Information:***
    - Data processed to reveal meaning.

What is a database?

- A Database consists of columns (attributes) and rows (records).
- Database Management System (**DBMS**)
  - –add, remove, update records
  - –retrieve data that match certain criteria
  - –cross-reference data in different tables
  - –perform complex aggregate calculations

Why Study Databases?

- They touch every aspect of our lives.
- Explosion of unstructured data on the web:
  - Large document collections.
  - Image databases, streaming media, social media
- Applications:
  - *Banking*: All transactions
  - *Airlines*: Reservations, schedules
  - *Universities*: Registration, grades
- *Sales*: Customers, products, purchases
- *Manufacturing*: Production, inventory, orders, supply chain
- *Human Resources*: Employee records, salaries, tax deductions
- *Telecommunications*: Subscribers, usage, routing
- *Computer Accounts*: Privileges, quotas, usage
- *Records*: Climate, stock market, library holdings

Databases versus spreadsheets

- Databases versus spreadsheets
  - Easier and more efficient manipulation of data
  - Avoid redundancy

Single table database

ISBN	Title	Author ID	Author	Publisher ID	Publisher	Publisher phone number	Price
23-5432	Data Analytics	3	Maria O'Neill	1	Tandis Publisher	555-12345	€ 45.00
23-5433	Data for beginners	43	Sonya Smith	2	YY Publishers	678-0000	€ 45.00
23-5434	Data Wrangling	7	Anna Redford	2	YY Publishers	678-0000	€ 32.00
23-5435	Data and marketing	52	Asha Fitzgerald	5	JK Ltd	543-3333	€ 65.00
43-2345	Data Visualisation	3	Maria O'Neill	1	Tandis Publisher	555-12345	€ 34.00
22-3932	Smart marketing	52	Asha Fitzgerald	5	JK Ltd	543-3333	€ 43.00
22-3933	Irish data centers	43	Sonya Smith	2	YY Publishers	678-0000	€ 44.00
43-2145	AI for beginners	12	Marek Schulz	5	JK Ltd	543-3333	€ 64.00
22-3938	Dublin art	3	Maria O'Neill	2	YY Publishers	678-0000	€ 80.00
45-2145	Machine learning	43	Sonya Smith	1	Tandis Publisher	555-12345	€ 78.00
45-2146	Advanced Data Visualisation	52	Asha Fitzgerald	1	Tandis Publisher	555-12345	€ 25.00
45-2147	Working with Data	7	Anna Redford	2	YY Publishers	678-0000	€ 26.00

Disadvantage of a single table database

- Redundancy of data
- Problem with complex data
- Problems in updating in bulk (new phone number)
- Problems in adding incomplete data (new publisher)
- Problems in removing group of data (all books from the publisher)

Several tables- Books

ISBN	Title	Author ID	Publisher ID	Price
23-5432	Data Analytics	3	1	€ 45.00
23-5433	Data for beginners	43	2	€ 45.00
23-5434	Data Wrangling	7	2	€ 32.00
23-5435	Data and marketing	52	5	€ 65.00
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Several tables- Authors

Author ID	Author
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Several tables- Publishers

Publisher ID	Publisher	Publisher phone number
1	Tandis Publishers	555-12345
2	YY Publishers	678-0000
2	YY Publishers	678-0000
5	JK Ltd	543-3333
1	Tandis Publishers	555-12345
5	JK Ltd	543-3333
2	YY Publishers	678-0000
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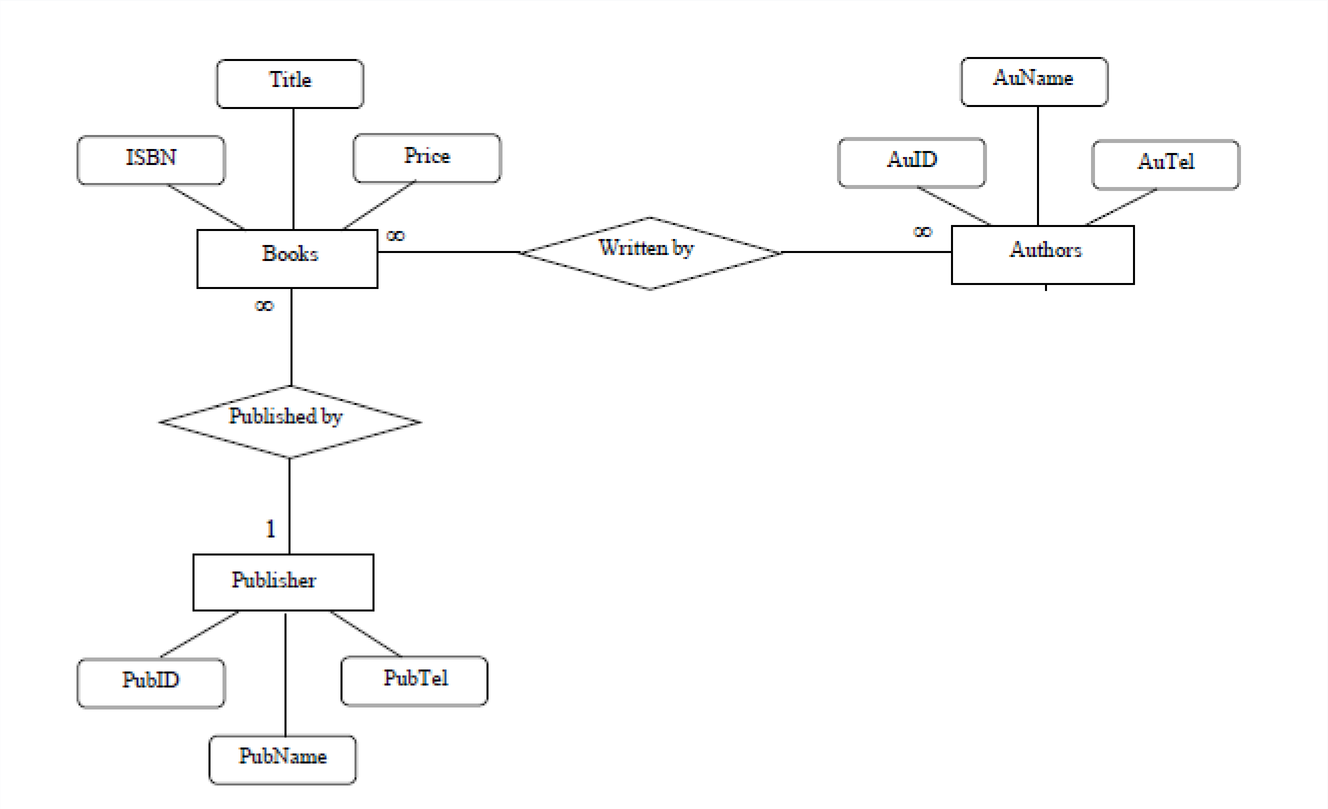
1	Tandis Publishers	555-12345
1	Tandis Publishers	555-12345
2	YY Publishers	678-0000

Publisher ID	**Publisher**	**Publisher phone number**
1	Tandis Publishers	555-12345
2	YY Publishers	678-0000
5	JK Ltd	543-3333

Relationships in a database example



Relationships

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22-3938	3
45-2145	43
45-2146	52
45-2147	7

Types of Databases

- **Can be classified by location…**
  - ***Centralized:***
    - Supports data located at a single site.
  - ***Distributed:***
    - Supports data distributed across several sites.

Types of Databases

- **Can be classified by use…**
  - **Transactional (or Production):**
    - Supports a company’s day-to-day operations.
  - **Data Warehouse:**
    - Stores data used to generate information required to make tactical or strategic decisions.
    - Often used to store historical data.
    - Structure is quite different.

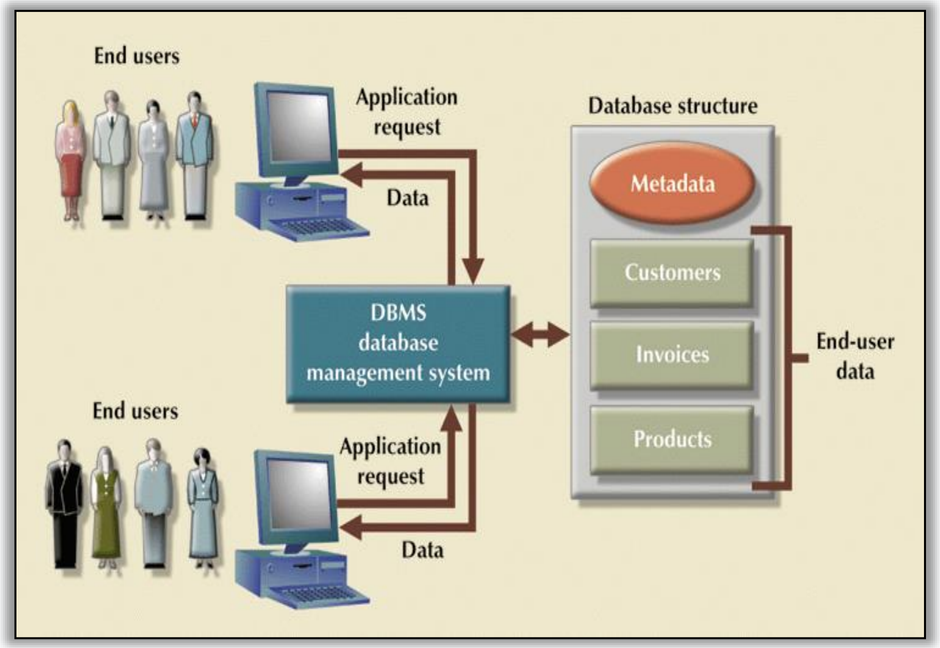
Types of Databases

- **Can be classified by relational model…**
  - ***Relational:***
    - SQL.
  - ***NonRelational:***
    - NonSQL.

Database Management Systems (DBMS)

- General purpose software for…
  - *definition*,
  - *construction*, and
  - *manipulation*

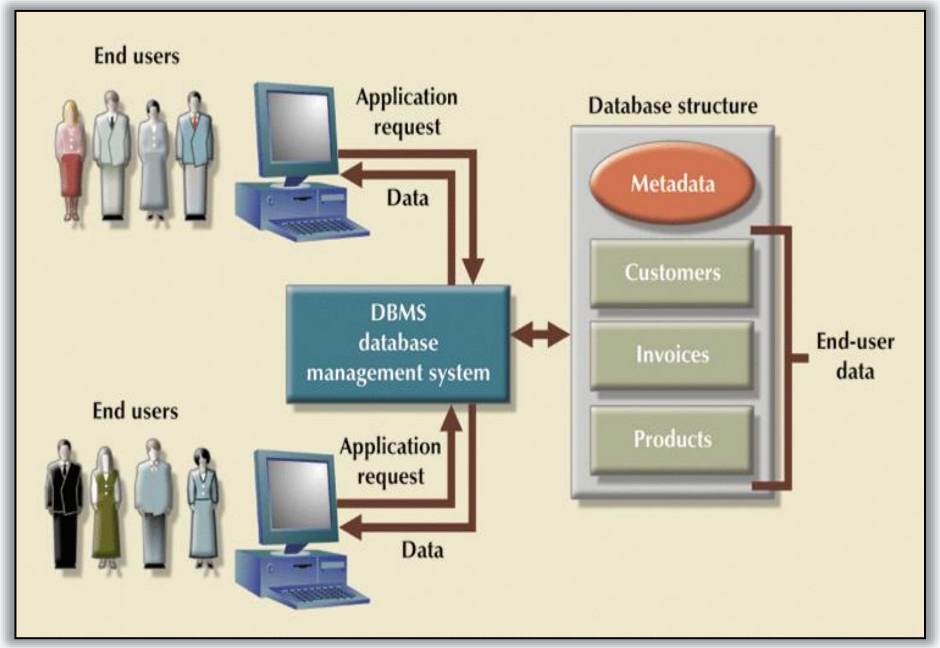
  of databases.



The DBMS manages the interaction between the end user and the database

Database Management Systems (DBMS)

- Characteristics of DBMS
  - Data is…
    - integrated
    - shared
    - persistent
    - self-describing
  - Abstraction
    - Program-data independence.
  - Allows multiple views of the data



The DBMS manages the interaction between the end user and the database

DBMS Functions

- DBMS performs functions that guarantee integrity and consistency of data:
  - **Backup & Recovery Management**
  - **Data Integrity Management**
  - **Database Access Languages & Application Programming Interfaces**
  - **Database Communication Interfaces**

Database Design

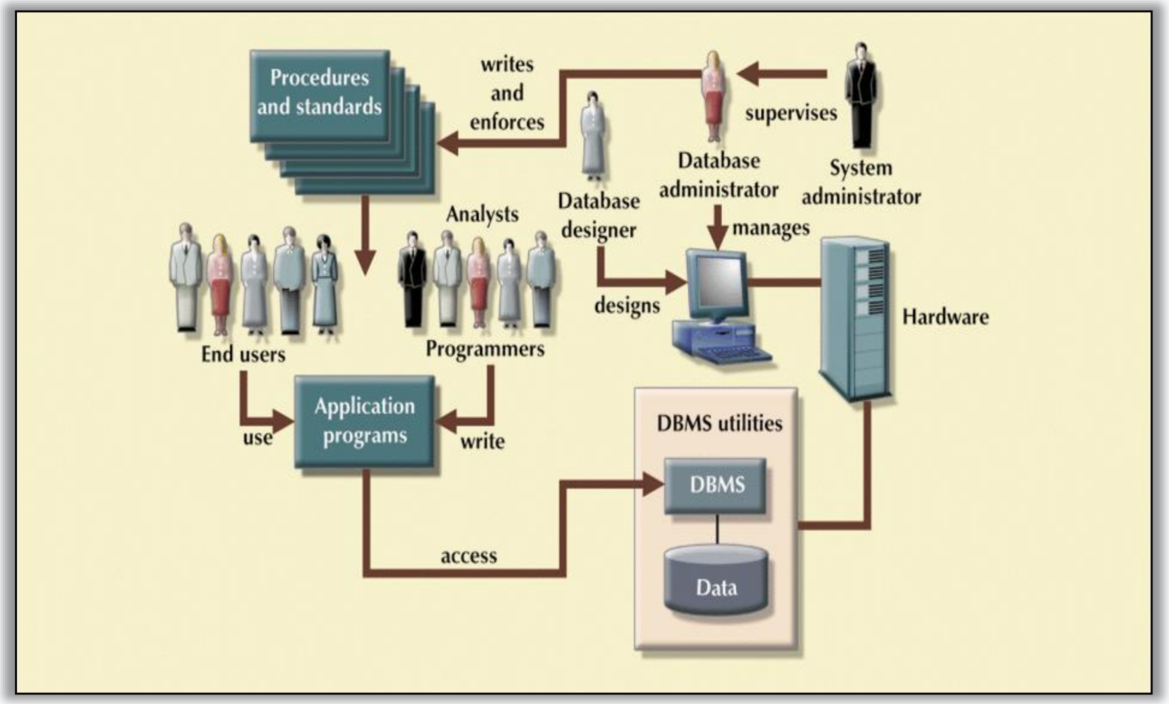
- Database design deals with how to design a database.
- Importance of good design:
  - Poor design results in unwanted ***data redundancy***.
  - Poor design generates ***errors*** leading to bad decisions.
- Practical approach:
  - Focus on principles and concepts of database design.
  - Importance of logical design.

Data Models

- Specify concepts that are used to describe the database. These can be categorised as:
    - ***High Level (Conceptual):***
    - ***Representational (Logical):***
    - ***Low Level (Physical):***

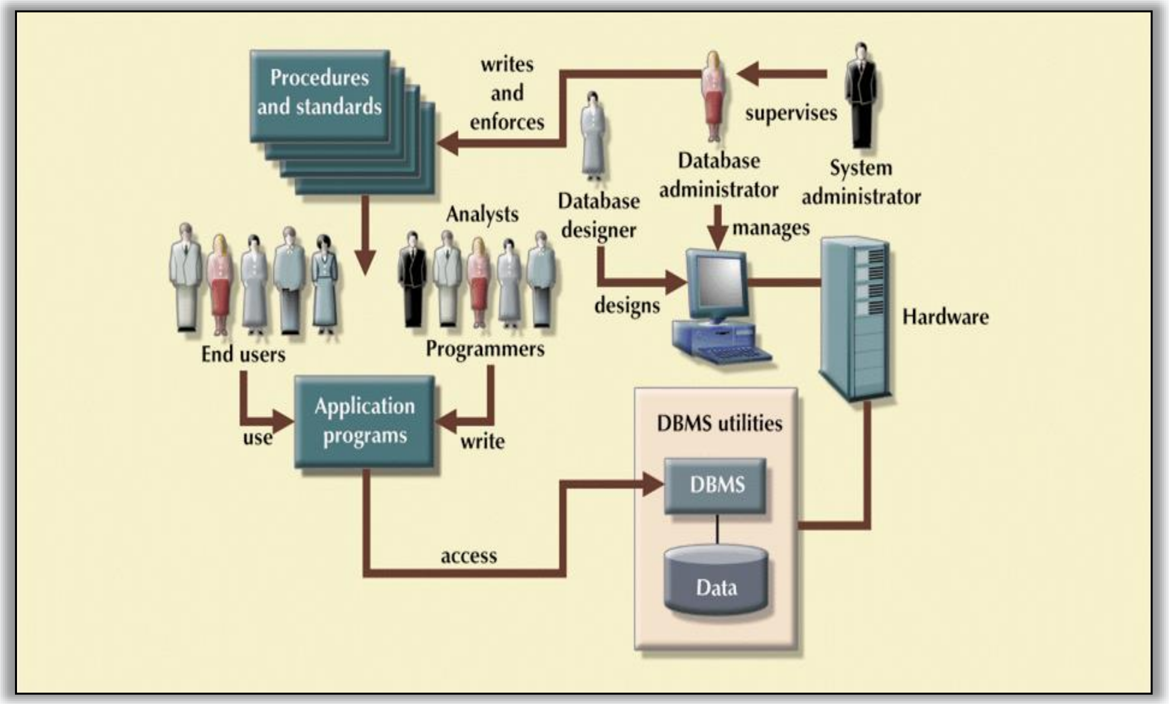
Roles in the Database Environment

- ***Database Administrator (DBA)***
  - Authorizing access to the database.
  - Coordinating and monitoring its use.
  - Acquiring software and hardware as needed.
  - Solving problems such as breach of security or poor system performance.



Roles in the Database Environment

- ***Database Designers/Modellers:***
  - Identifying the data to be stored and structure to be used, identify constraints.
  - Design structural and functional aspects.
- ***Application Programmers:***
  - Develop and implement end-user requirements.
- ***End Users:***
  - Access to the database: querying, updating, generating reports.



Software

- MS Access
- Microsoft SQL Server
- Oracle RDBMS
- SAP Sybase ASE
- IBM DB2
- Amazon RDS
- MySQL
- SQLite
- PostgresSQL
- MongoDB
- Redis
- Cassandra
- Neo4j
- phpMyAdmin

Software

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- Cassandra
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- phpMyAdmin

Summary

- What is a database
- Why use databases (Databases vs. Spreasheets)
- Types of Databases
- DBMS
- Database Design and models
- Software

Thanks

- Edmund Nevin and Marek Kręglewski for a earlier versions of these lecture notes

Lab 1

- **Getting familiar with the lab set up**

- **Installing the required software**

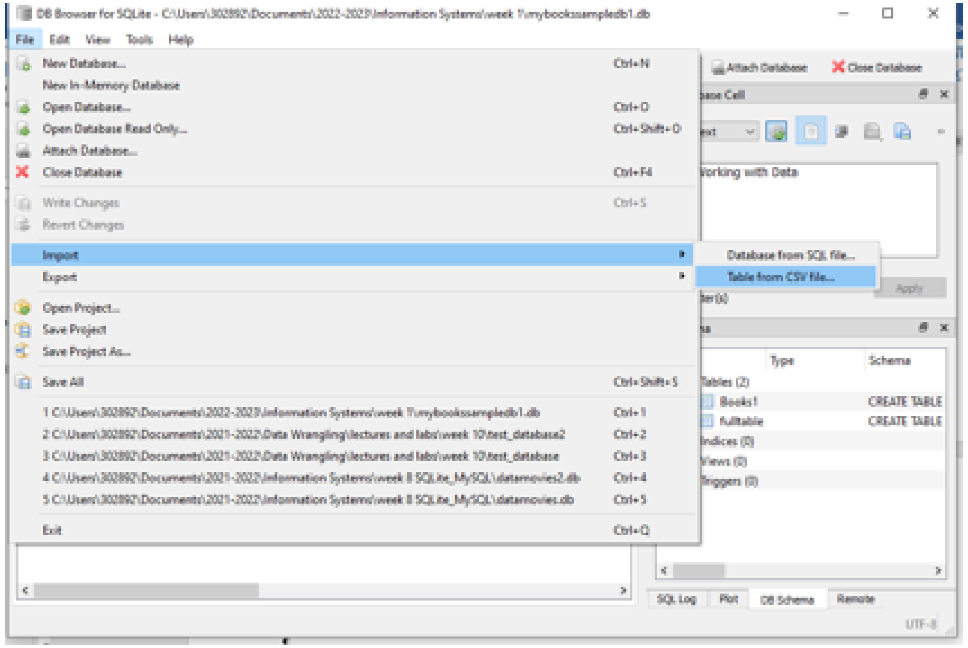
https://sqlitebrowser.org

Lab 1

Sample Database

Create a New Database called sample1.

Import Books1.csv



Identify the redundancy level and update efforts required to, for example, change a publisher's phone number. Use the Tab Browse Data to view and update data.

Break
Back at 8pm
for Lab session

TU257
Information Systems
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